

ASSIGNMENT 2

Internal Service Agent — IT Support

Veridian Corp Enterprise Support Prototype · September 2026





Problem Statement & Scope

Corporate Function

Internal IT Support Agent for Veridian Corp employees

Operating Context

Week of September 21–25, 2026

Primary Goal

Convert raw employee requests into accurate resolutions, structured tickets, and policy citations

Key Requirement

Strict compliance with KB-01 through KB-10 — zero policy hallucination

Agent Architecture & Workflow

A five-stage pipeline transforms unstructured employee requests into auditable, policy-grounded outcomes.



Each stage enforces policy compliance before passing control to the next, ensuring no request bypasses KB grounding.

Knowledge Base Grounding & Policy Mapping

Article / Policy	Operational Scope	Policy Constraint / Trigger
KB-01: Password & Account Access	Account resets & lockouts	Self-service allowed up to 5 attempts; > 5 requires manual IT unlock.
KB-03: Hardware Replacements	Laptop/desktop lifecycle	Standard 4-year cycle; early replacement permitted after 3 years for complete hardware failure.
KB-07: Guest Wi-Fi Access	Temporary network access	Self-service generation via front-desk kiosk (valid for 24 hours).
KB-09: Security Incident Reporting	Threat mitigation	Immediate halt on forwarding suspicious emails; auto-escalate to Security Ops.
KB-10: Work From Home Equipment	Remote setup requests	Requires dual-manager approval + HR remote-work validation.

Core Decision Engine Logic



Direct Resolution

Standardized, self-service requests (e.g., KB-07 Guest Wi-Fi kiosk)



Clarification Loop

Triggered by vague or incomplete inputs (e.g., REQ-15 "it's not working")



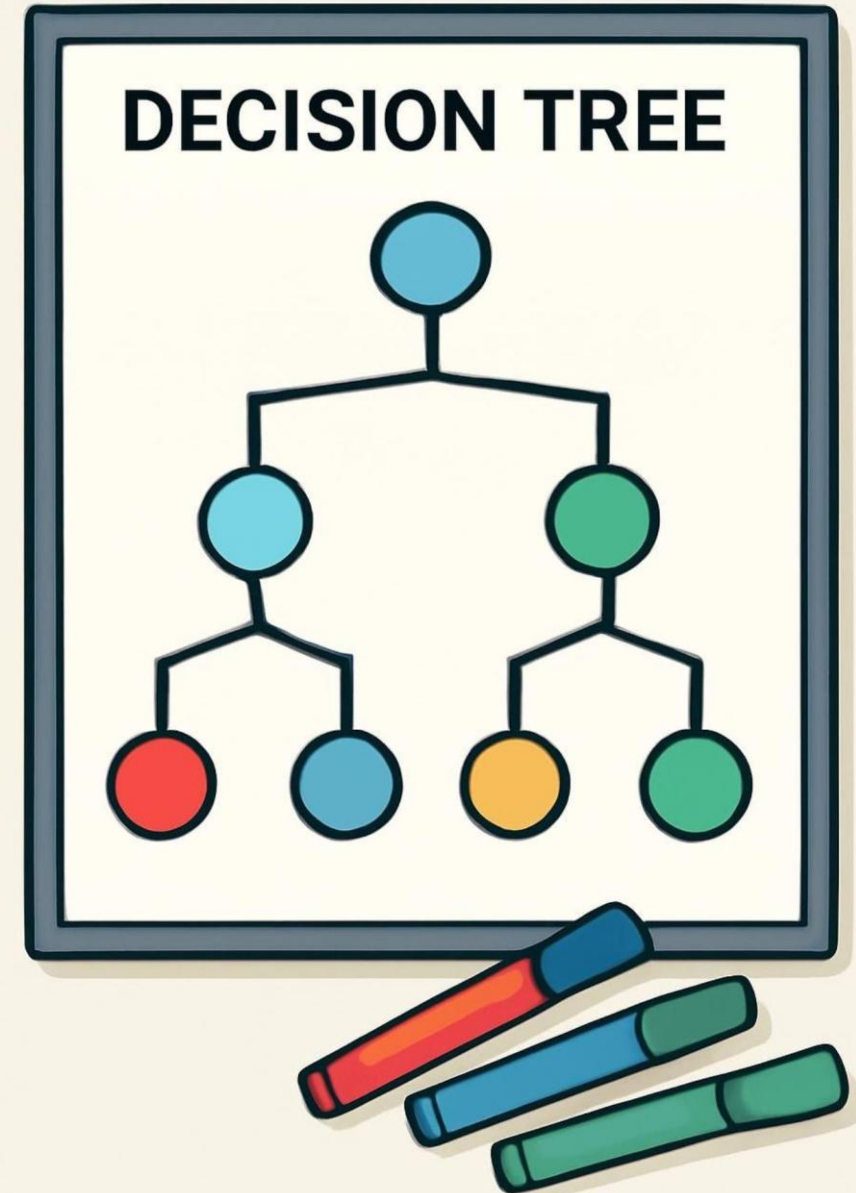
Manual Action

Queues tickets for IT administration (e.g., REQ-03 locked accounts)



Approval Path

Routes multi-department sign-off requests (e.g., KB-10 WFH equipment)



Security Interception & Risk Mitigation



Phishing Interception

Immediately halts email forwarding upon threat detection (REQ-08). Prevents lateral spread of malicious content.



Access Control Enforcement

Strict authorization policies for admin server access, following historical precedent (TK-1050).



Policy Guardrails

All outputs grounded in verifiable KB sources — invented resolutions are structurally impossible.

Structured Ticket Schema & Audit Logging

Security Guardrails & Threat Interception

- **Rule #1: Immediate Action Blocking**
 - Detects risky user intentions (e.g., forwarding active phishing links or sharing credentials) and halts execution before damage occurs.
- **Rule #2: Zero-Trust Threat Isolation**
 - Threat inputs bypass normal queueing and are routed directly to Security Operations under KB-09.
- **Rule #3: Clear Safety Instructions**
 - Provides employees with explicit, direct instructions (e.g., *"Do not forward or click links in this email"*).
- **Rule #4: High-Severity Audit Logging**
 - Flags the ticket status as Escalated to Security with high severity in the central audit ledger for rapid SOC response.

Test Benchmark: Request Pipeline Performance

1

REQ-01 · Aditi Sharma

Dead laptop (3.5 yrs) → **Approved**
under KB-03 3-year threshold

2

REQ-02 · Vikram Chawla

Guest Wi-Fi → **Directed** to front-
desk kiosk per KB-07

3

REQ-08 · Ananya Reddy

Phishing email → **Intercepted &
routed** to security per KB-09

4

REQ-12 · Sneha Kulkarni

Expense tool login → **Clarified** account status per KB-08

5

REQ-15 · Rahul Menon

Ambiguous prompt → **Automated follow-up** triggered

AI Stack & Implementation Tools



Frontend / UI

Streamlit for rapid web prototyping and interactive agent testing



Logic & Orchestration

Python & System Prompting for deterministic policy enforcement



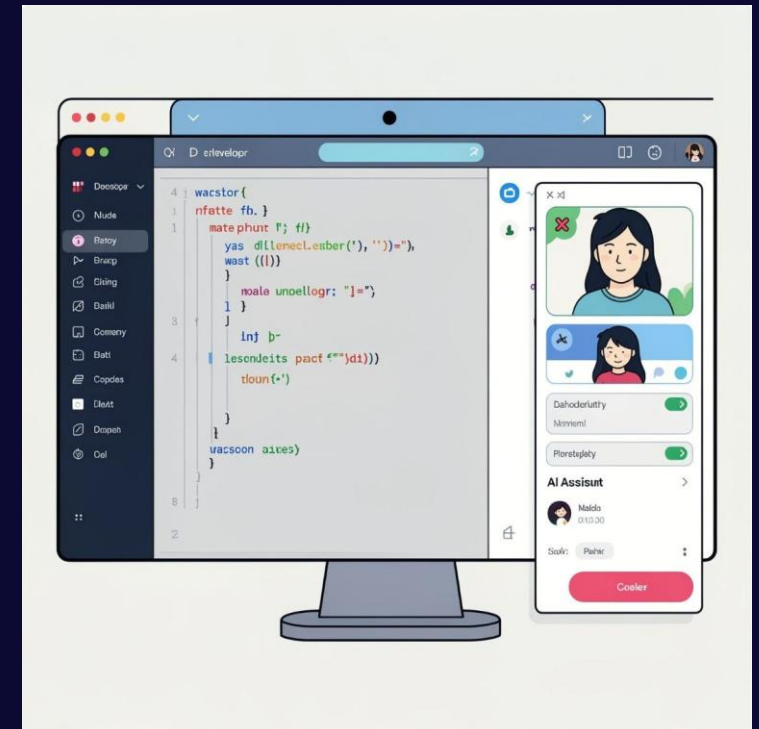
Version Control & Hosting

GitHub Repository and Streamlit Community Cloud deployment



AI Assistive Tools

ChatGPT for logic validation



Future Roadmap & Enterprise Scaling

1

ITSM Integration

Native sync with ServiceNow and Jira Service Desk for bi-directional ticket management

2

Omnichannel Support

Deploy across Slack, Microsoft Teams, and email webhooks for unified employee access

3

Advanced SLA Monitoring

Real-time tracking of resolution times, escalation thresholds, and compliance metrics

Each roadmap phase builds on the existing KB-grounded architecture — no policy compliance is sacrificed for scale.

ENTERPRISE TECHNOLOGY ROADMAP

